

STUDY AND EVALUATION SCHEME

Of

Diploma (Engineering)- 3Years

Diploma (Engineering)- Civil Engineering III Year

Semester -V



**BHABHA UNIVERSITY,
BHOPAL (M.P.)**



Bhabha University, Bhopal (M.P.)

Scheme of Examination
Programme – Three Years Diploma in Civil Engineering

Semester -V

Branch – Civil Engineering

Subject Code	Subject	Theory Component								Practical Component						Total Credit	Grand Total of Marks
		LECTURES	CONTINUOUS EVALUATION			END OF THE TERM/ SEMESTER EVALUATION			THEORY CREDIT	PRACTICAL Hrs. Per Week	CONTINUOUS EVALUATION	END OF THE TERM/ SEMESTER EVALUATION			PRACTICAL CREDIT		
			Hrs. Per Week	TERM WORK, QUIZ, ASSIGNMENT	MID TERM TEST (TWO)	THEORY PAPER						LAB WORK, QUIZ, ASSIGNMENT	PRACTIAL / ORAL EXAMINATION (VIVA)				
NO.	MARKS	DURATI ON (Hr.)				NO.	MARKS	DURATI ON (Hr.)									
CE 5001	Irrigation Engineering	05	10	10	10	01	70	3 hrs	04			1				04	100
CE 5002	Q.S.C.-I	06	10	10	10	01	70	3 hrs	04			1				04	100
CE 5003	Work organization & management	06	10	10	10	01	70	3 hrs	04			1				04	100
CE 5004	Transportation Engineering II	04	10	10	10	01	70	3 hrs	04	02	20	1	30	3 hrs	2	06	150
CE 5005	S.D.D.-I (RCC)	06	10	10	10	01	70	3 hrs	04	02	20	1	30	3 hrs	2	06	150
CE 5006	Field visit									03	40		60	3 hrs	2	02	100
	TOTAL	27	50	50	50	05	350		20	7	80	05	120		6	26	700

Theory Credits : 20

Practical Credits : 06

Total Credits : 26

Theory Mark : 350

Practical Marks : 120

Quiz, Mid Term, Lab. Work : 230

Minimum Pass Grade in Theory & Practical 'D'

BHABHA UNIVERSITY, BHOPAL

Subject – Irrigation Engineering

Sub Code – CE5001

UNIT 1

Introduction

Definition – Irrigation and irrigation Engineering, advantages of irrigation, ill effects of over irrigation, and types of irrigation project purpose wise and administrative wise, Methods of irrigation. Analyze data for irrigation project, supervision of reservoir and canal structure, weir and barrages, lift irrigation scheme, its suitability, advantages and limitations Capacity of reservoir, Principle of Hydrology Relation between water and crop Rainfall, Crops, Dams Weir, Barrages, Area Capacity curve Capacity Canal Concept of runoff duty delta and base period

UNIT 2

Hydrology

Hydrological cycle, Definition of rainfall , raingauge and rain gauge station , types of rain gauges (names only) average annual rain falland its calculation , definition of run off, factor affecting run off, calculation of run off by run offcoefficient, English formula , Stranges and Binnie's tables and curves. Maximum flooddisharge and methods of calculation. Unit hydrograph Yield and Dependable yield andmethods calculation.

Water Requirement crops

Cropping seasons and crop in Madhya Pradesh. Definition – Crop period base period Duty Delta ,factors affecting Duty , relation between Duty Delta and base period Definition – CCA , GCA,IA, intensity of irrigation time factor capacity factor. Crop rotation. Problems on water requirement and capacity of canal. Assessment of irrigation water

UNIT 3

Investigation and reservoir planning

Survey for irrigation project data collected for irrigation project. Area capacity curve, silting of reservoir, rate of silting, factors affecting silting, methods to control levels and respective storage in reservoir. Fixing control levels

DAMS AND SPILLWAYS

Types of dams – Earthen dams and Gravitydams (masonry and concrete) Comparison of earthen and gravity dams with respect to foundation, seepage, construction and maintenance Earthen Dams – Components and their function , typical cross section seepage through embankment and foundation seepage control through embankment and foundation .

Methods of constructions, types of failure of earthen dams and remedial measures. GravityDams Theoretical and practical profile, typical cross section, drainage gallery, joint in gravitydam, high dam and low dam Spillways-Definition,function, location and components. Emergency and services, ogee spillway and bar type spillway, discharge over spillway. Spillway with and with out gates

UNIT 4

Small irrigation structure ,bandhara, percolation tanks and lift irrigation

Advantages and disadvantages of Bandharairrigation layout and component parts, solid and open Bandhara. Percolation Tanks – necessity and importance, selection of site. Layout of lift irrigation scheme. Irrigation department standard design and specification. Small irrigation structures, like Stop dam, stop dam cum causeway, ring bund , small ponds

UNIT 5

Diversion Head Works

Weirs – components parts, function and types, layout of diversion head works with its components and their function, canal head regulator, silt excluders and silt ejectors.

Barrages

– components and their function. Difference between weir and barrage irrigation department standard design and specifications.

CANALS

Classification of canals according to alignment and position in the canal network. Design of most economical canal section. Canal lining –Definition, purpose, types of canal lining advantages of canal lining properties of good canal lining material. CD. works- different C.D. works, canal falls, escapes, cross regulators and canal outlets

LIST OF REFERENCE BOOKS

S.No.	Title	Author	Publisher
1	Irrigation and water power Engineering	B.C. Punmia	
2.	Introductory Irrigation Engineering	B.C. Punmia	Laxmi Publication, Delhi.
3	Fundamental principle of Irrigation Engineering	V.B.Priyani	
4	Fundamental principles of Irrigation Engineering	Bharat Singh	

BHABHA UNIVERSITY, BHOPPAL

Subject – Quantitative surveying estimating & costing – I

Sub Code – CE5002

UNIT -1

Overview Of Estimating & Costing

Meaning of the terms estimating, costing. Purpose of estimating and costing .Types of estimate - Approximate and Detailed. Approximate estimate Types- Plinth area rate method, Cubic Content method, Service Unit method, Typical bay method, Approximate Quantity method , Problems on Plinth area rate method & application of Service unit method for selection of service unit for different types of civil Engineering Structures. Types of detailed estimate Detailed estimate for newwork. Revised estimate. Supplementary estimate. Revised & Supplementary estimate. Maintenance & Repair estimate. Uses of detailed estimate

UNIT -2

Detailed Estimate

Unit quantity method, Total quantity method, Data required for detailed estimate. Factors to be considered during preparation of detailed estimate, Specification, Quantity availability of material, Location of site, Labour Component. Steps in preparing detailed estimate. Taking out quantities, squaring, abstracting. Preparing check list – by adoption of Sequence of execution. drafting Brief Specification of items, contents of measurement Sheet , Abstract sheet, face sheet

UNIT -3

Mode of Measurements

General Rules for fixing units of Measurements for different– items of work as per IS 1200 & As per PWD Hand Book Desired accuracy in taking measurements of various items of work & rules for deductions as per IS 1200 & P.W.D. handbook

UNIT -4

Procedure for Preparing Detailed Estimat

Procedure for taking out quantities for various items of works by P.W.D & IS 1200 for. a) for Load bearing Structure –Long Wall and short wall method , Center line method . b) Framed Structure building. --- By using thumb rules for reinforcement quantity calculation - By preparing bar bending Schedule Provisions in detailed estimate for contingencies, work charged establishment, Provisional items, Provisional Sum, Provision for water Supply & Sanitary works, Electrical wiring &installations, centage charges, Tools &Plants, Prime cost, Day work

UNIT -5

Rate analysis

Meaning of term Rate analysis –Factors affecting rate analysis, lead, lift, task work, materials and labourcomponent, Market Rate and labour rate. Transportation of Materials, load factor for different materials. Standard lead , extra lead, Transportation Charges , Labour - Categories of labours, labour rates, overheads contractor's profit, water charges, taking out quantities of materials for different items of works. Preparing rate analysis of different items of work Standard Schedule of rates, full rates &labour rates

Taking out quantities of work for different Civil Engineering Works

Roads, Dam , Canals ,Railway embankments, methods of mean area , mid sectional area, trapezoidal, Prismoidalformula. Calculation of quantity of earth work. Estimate of Road of 1km. length for pavement surface WBM Bitumen Cement concrete road Use of software for estimation & for analysis of rates

LIST OF REFERENCE BOOKS

S.No.	Title	Author	Publisher
01	Estimating & costing in Civil Engineering	B.N. Datta	UBS Publishers Distributors Pvt Ltd New Delhi
02	Estimating & costing, Specification and Valuation in Civil Engineering	M. Chakraborti M. Chakraborti , Calcutta	
03	Estimating & costing	S.C. Rangwala Charotar	Publication Anand

List of Assignments:

- 1) Prepare Check list of items of following type of Civil Engineering works.
 - a) Load Bearing type Building
 - b) Framed structure type building
 - c) W.B.M.Road
 - d) Septic Tank
 - e) Community well
- 2) Writing the rules of deduction's for below mentioned items of work as per IS 1200.
 - a) Brick / Stone masonry.
 - b) Plastering / Pointing
- 3) Taking out quantities of various items of work for load bearing building.
 - i) Earth work in excavation for foundation
 - ii) Base Concrete of foundation
 - iii) U.C.R. /BB Masonry work in foundation and plinth.
 - iv) D.P.C.
 - v) Plinth Filling.
 - vi) Brick work in masonry.

vii) Flooring

viii) Plastering.

ix) Wood work in doors & windows

4) Taking out quantities of following items for small R.C.C. Hall

i) Concreting for footing, Column, Beam, slab.

ii) Reinforcement for above items by preparing Schedule of bars.

iii) Form work for all above items.

5) Preparing detailed estimate of a RCC single & two storied residential building for all items

of work. (The quantity of reinforcement shall be calculated by percentage.)

6) Preparing Rate analysis of following items:

Building work – Brick work, P.C.C., R.C.C., Plastering, Flooring, Doors, Windows..

BHABHA UNIVERSITY, BHOPPAL
Subject – Work organization & management
Sub Code – CE5002

Unit -1

Procedure of Execution of work by P.W.D. Organization of P.W.D. functions of their personnel. P.W.D. procedure of initiating the work administrative approval, technical sanction, budget provision. Method used in P.W.D. for carrying out works contract method and departmental method, Rate list method, piece work method, day's work method, department method.

(NMR and casual muster roll)

Unit -2

Contract

Definition of contract, objects of contract, requirements of valid contract. Types of engineering contract- Lump sum contract, item rate contract, percentage rate contract, cost plus percentage, cost plus fixed fee, cost plus variable percentage and cost plus variable fee contract, labor contract, demolition contract, fee contract, target contract, negotiated contract. Class of contractor, Registration of contractor. BOT Project.

Unit -3

Tender & Tender documents

Definition of Tender, necessity of Tender, Types of Local & Global. Tender Notice, points to be included while drafting Tender Notice, Drafting of Tender Notice. Meaning of terms: Earnest money, security deposit, validity period, right to reject one or all tenders, corrigendum to tender notice and its necessary. Tender documents – List, scheduled A, Schedule B, Schedule C. Terms related to Tender documents – Contract conditions, time limit, time extension, penalty, defective material and workmanship, Termination of contract, Suspension of work, subletting of contract, extra item, escalation, arbitration, price variation clause, defect liability

period, liquidated and un liquidated damages. Filling the tender by contractor and points to be observed by him. Procedure of submitting filled in Tender document. Procedure of opening tender, comparative statement, scrutiny of tenders, award of contract, acceptance letter and work order. Unbalanced Tender, Ring Formation

Unit -4

Accounts of P.W.D

Various Accounts Forms and their uses – measurement, Books, Nominal Muster Roll, Imprest Cash, indent, Invoice, Bills, Vouchers, Cash Book, Temporary advance

Payment to Contractors

Mode of payment to the contractor : Interim payment and its necessity, Advance payment, secured advance, on account payment, Final payment, first and final payment, retention money, reduced rate payment, petty advance, mobilization advance

Specifications

Necessity and importance of specifications of an items, points to be observed in framing specifications of an item, types of specification. Brief and detailed, standard and

manufacturers specifications. Preparing detailed specifications of items in Civil engineering works, standards specification book. Legal aspects of specification

Chapter 5

Cash, Bills, Auction & T.A. rules

Procedure to settle account of money received, modes of payment, permanent and temporary advance, comparison, checking of bills and vouchers, auction procedure, T.A. rules etc.

Time schedule for works

Importance of management of works Gantt bar chart, limitation of chart, CPM network, project chart

Miscellaneous

Necessity of maintaining daily dairy, need for presence of sub engineer, A/R & S/R , charge to be handled to be cash transferred, inspection of rest houses. Measures to improve the efficiency of labour, causes of accident, trade unions, aims of labour legislation, labour courts, attitudes of sectional officers towards labour

Assignments :

1. Collecting old set of tender document and writing a report on it.
2. Collection of tender notices published in newspapers for various items of civil engineering works (At least 5) write salient features of them.
3. Drafting Tender Notice for construction of a Civil Engineering work (W.B.M. Road, Residential Building)
4. Preparation of Tender Document for the building. (Detailed Estimate prepared for R.C.C. building in estimating and costing shall be used)
5. Collection of various account forms from PWD & wiring report on in it.
6. Writing a report on store procedure and account producer of PWD. For it A –
 - a) Guest Lecture of PWD Official may be arranged.
7. Writing detailed specifications for one item from each of following :
 - A) Building construction system.
 - B) Irrigation engineering system.
 - C) Transportation engineering system.
 - D) Environment engineering system.
8. Preparing muster rolls.
9. Preparing imprest account and temporary advance forms and developing skill for filling in forms.
10. Solving CPM and Net work problems
11. CPM PERT RELATED SOFTWARE
12. Preparation a ‘ E’ Tendering of a particular project .

VISITS :

1. Visit to public sector/Govt. Industry/ Organization.like PWD ,RES,
2. Visit to private sector Industry

REFERENCE BOOKS :

S.No.	Title	Author	Publisher
01	A.B.C. of PWD Accounts	C.M. Kaul	
02	Overseer accounts & Duties	Kumar	
03	PWD Managements, Accounts &LabourRelation	H.S. Pandit	
04	Construction Management &PWDAccounts	Agrawal & Arora	
05	MPPWD Manual Vol-I & Vol-II		
06	Manual of Labour Relations	R.C. Shrivastava	

BHABHA UNIVERSITY, BHOPPAL

Subject – Transportation Engineering II

Sub Code – CE5004

Unit -1

Road Engineering

Importance of road in India. Classification of roads according to Nagpur plan (Location and function), and third road development plan. Traffic and tonnage, Classification of urban roads. different road yojana ,like pradhan mantra gram sadakyojana ,Mukhya mantra sadakyojna

Unit -2

Investigation for road project

Reconnaissance survey, Preliminary survey and Location survey for a road project. Detailed survey for cross drainage- L-section and C/S sections. Fixing the alignment of road, factors affecting alignment of road. Drawings required for road project- Key map, Index map, Preliminary survey plan and detailed location survey plan, L section and C/S sections cross drainage work, land acquisition plan. Survey for availability of construction material, location plan of quarrie.

Unit -3

Geometric design of highways

Camber- definition, purpose, types, IRC – specifications. Kerbs, road margin, road formation, right of way. Design speed- IRC – specifications. Gradient – definition, types, IRC specification. Sight distances– definition, types, IRC specification. Curves–Necessity, types– horizontal, vertical and transition curves. Widening of roads on curves. Super Elevation – definition, formula for calculating super elevation, minimum and maximum values of super elevation, and methods of providing super elevation. Sketching of standard C/S of national highway in embankment and cutting. Simple problems on geometric design of road

Unit -4

Construction of roadspavements and materials

Types of road materials and Tests – soil, aggregates, bitumen, Cement Concrete. Test on soil sub grade- C.B.R. test, Test on Aggregate – Los Angeles abrasion, impact, and shape test. Tests on bitumen Penetration, Ductility and Softening point test. Pavement – objective of pavement, structure of pavement, function of pavementcomponents, types of pavement. Construction of earthen road – general terms used- borrows pits, spoil bank, lead and lift, balancing of earthwork. Construction procedure. Soil stabilized roads – necessity, methods of soil stabilization, brief details of mechanical soil stabilization. Water bound macadam roads – materials used, size and grading of aggregates and screening, construction procedure including precautions in rolling. Construction of bituminous roads.Terms used– bitumen, asphalt, emulsion, cutback, tar, common grades adopted for construction. Types of bituminous surface – prime coat, tack coat, seal coat, Surface dressing – procedure of construction bituminous penetration macadam, and Bitumen/Tar carpets – procedure of construction. Cement concrete pavements Construction procedure and equipments, Construction joints, joint filler, joint sealer.

Unit -5

Traffic engineering

Traffic volume study, Traffic control devices road signs, marking, Signals, Traffic Island. Road intersections- intersections at grade and grade separator intersections. Road accident. Building code IS:1904. Definition of active earth pressure and passive earth pressure, structures subjected to earth pressure in the field

Hill roads:

Parts and functions of hill road components, types of curves, Hill road formation. Land slides- causes and prevention. Structures drainage structures

Drainage of roads and maintenance and repairs of roads

Surface drainage – side gutter, catch water drains, surface drainage. Sub-surface

drainage –Longitudinal drains and cross drains

Necessity of maintenance of roads,

Classification of maintenance operation – ordinary, routine and periodic maintenance.

Maintenance of W.B.M., bituminous and cement concrete roads.

Arboriculture

Road side arboriculture, necessity, planning of plantation of trees selection of types of trees and development of nursery considering the environment aspects

REFERENCE BOOKS:

S.No.	Title	Author	Publisher
01	Highway Engineering	Khanna & Justo	Khanna Pub.
02	Traffic Engineering	L.R. Kadiyali	
03	Transportation Engineering	N.L.Arora,S.P.Luthara	I.P.H. New Delhi
04	Transportation Engineering	Vazarani&Chandola	Khanna Pub.
05	Road, Railway, Bridges	Biridi& Ahuja.	S.B.H.New Delhi
06	Transportation Engineering	Kamala	T.M.H. New Delhi

BHABHA UNIVERSITY, BHOPPAL

Subject – Structural Design & Drafting – I (RCC)

Sub Code – CE5005

Unit -1

Introduction to RCC

S.I. Units, Meaning of R.C.C. purpose of reinforcement. Materials of reinforcement steel as a reinforcing material. Types of steel used for reinforcement mild steel, Tor steel, permissible stresses in concrete and steel. Different mixes of concrete to be used for R.C.C. work use of I.S. code No. 456-2000 and I.S. 875-1984 for designing R.C.C. structures. Introduction to RCC design software like STRUUDS, resist

Fixed & continuous beam

Concept of fixity, effect of fixity, advantages and disadvantages of fixed beam. Fixed end moments from first principle for beam subjected to UDL over entire span, central point load, Point load other than mid span. Application of standard formulae in finding moments and drawing S.F. and B.M. diagrams for a fixed beam. Clapeyron's theorem of three moment (no derivation). Application of theorem maximum up to three spans and two unknown support moment only, Support at same level, spans having same moment of inertia subjected to concentrated loads and uniformly distributed loads over entire span. Drawing SF and BM diagrams for continuous beams.

Unit -2

Working Stress Method & Prestressed Concrete.

Introduction to reinforced concrete, R.C. Sections their behavior, grades of concrete steel. Permissible stresses, Assumptions in W.S.M. Equivalent bending stress distribution diagram for singly reinforced section Concept of under-reinforced, over-reinforced and balanced section, neutral axis co-efficient Simple numerical problems on determining design constants, moment of resistance and area of steel for singly & doubly reinforced beam. Concept of pre stressed concrete, externally and internally

pre stressed member. Advantages and disadvantages of pre stressed concrete. Methods of pre stressing, pre tensioning and post tensioning. Losses in prestressing

Limit State Method

Definition, types of limit states, partial safety factors for materials strength, characteristics strength, characteristics load, design load. Loading on structure as per I.S. 875. I.S. Specification regarding spacing of reinforcement in slab, cover to reinforcement in slab, beam column & footing, minimum reinforcement in slab, beam & column, lapping, anchoring effective span for beam & slab.

Unit -3

Analysis and Design of Singly Reinforced Sections (LSM)

Limit State of collapse (Flexure), Assumptions stress. Strain relationship for concrete and steel neutral axis, Stress block diagram and Strain diagram for singly reinforced section. Concept of under-reinforced, over-reinforced and balanced section, neutral axis co-efficient, limiting value of moment of resistance and limiting percentage of steel required for balanced singly R.C. Section. Simple numerical problems on determining design constants, moment of resistance and area of steel.

Analysis and Design of Doubly Reinforced Sections (LSM)

General features, necessity of providing doubly reinforced section reinforcement limitations.

Analysis of doubly reinforced section, strain diagram stress diagram, depth of neutral axis, moment of resistance of the section. Simple numerical problems on finding moment of resistance and design of beam sections

Unit -4

Shear, Bond and Development Length (LSM)

Nominal Shear stress in R.C. Section, design shear strength of concrete, Maximum shear stress, Design of shear reinforcement, Minimum shear reinforcement, forms of shear reinforcement. Bond and types of bond, Bond Stress, check for bond stress,

Development length in tension and compression, anchorage value of hooks 90° bend and 45° bend Standard Lapping of bars, check for development length Simple numerical problems on deciding whether shear reinforcement is required or not, check for adequacy of the section in shear. Design of shear reinforcement; Minimum shear reinforcement in beams; Determination of Development length required for tension reinforcement of cantilever beams and slab, check for development length.

Unit -5

Analysis and Design of T-Beam (LSM)

General features, advantages, effective width of flange as per IS:456-2000 code provisions.

Analysis of singly reinforced T-Beam, strain diagram & stress diagram, depth of neutral axis, moment of resistance of T-beam Section with neutral axis lying within the flange. Design of T-beam for moment and shear for Neutral axis within or up to flange bottom. Simple numerical problems on deciding effective flange width. (Problems only on finding moment of resistance of T-beam section with N.A. lies within or up to the bottom of flange shall be asked in written examination.

Design of Slab (LSM)

Design of simply supported one-way slabs for flexure check for deflection control, and shear. Design of one-way cantilever slabs and cantilever chajjas for flexure check for deflection control and check for development length and shear.

Design of two-way simply supported slab for flexure with corner free to lift. Design of dog-legged staircase. Simple numerical problems on design of one-way simply supported slabs cantilever slab & two –way simply supported slab.

Design of Axially Loaded Column and Footing (LSM)

Assumptions in limit state of collapse- compression. Definition and classification of columns, effective length of column. Specification for minimum reinforcement; cover, maximum reinforcement, number of bars in rectangular, square and circular sections, diameter and spacing of lateral ties. Analysis and design of axially loaded short,

square; rectangular and circular columns with lateral ties only, check for short column and check for minimum eccentricity may be applied. Types of footing, Design of isolated square footing for flexure and shear. Simple numerical problems on the design of axially loaded short columns and isolated square footing.

REFERENCE BOOKS

S.No.	Authors	Title	Publisher
01	Dr. V.L. Shah & Late Dr. S.R. Karve.	Limit State Theory & Design of Reinforced Concrete.	Structure Publications
02	N.C. Sihna& S.K. Roy	Fundamentals of Reinforced concrete.	S.Chand& Company
03	N.Krishna Raju R.N. Pranesh	Reinforced concrete Design (IS 456-2000) Principles &Practice	New Age International
04	N. Krishna Raju	Prestressed Concrete	